# Section 5: Synthesis — Proof of the Main Theorem

This section assembles the four independently proved conclusions C1–C4 into a proof of the Main Theorem (Theorem 1 of the problem statement).

## Recalled Conclusions

- **C1 (Capability Inevitability, proved in Section 1):** Under A1, the scaling-law AI system exhibits exponentially decreasing loss, accelerating capability growth ($\kappa(t) = e^{\beta t}/A$ is strictly convex), and economically dominates the pre-AI baseline in every period.

- **C2 (Economic Lock-In, proved in Section 2):** Under A2 and A4, for any organization at integration depth $d \geq 1$ and any discount rate $\rho > 0$:

$$V_{\text{rev}}(d, \rho) - V_{\text{cont}}(d, \rho) \;\leq\; -c\, d^2 - \frac{\gamma \Pi_0}{\rho} \;<\; 0.$$

  Reversion is never individually economically rational; the case against reversion strengthens monotonically with depth.

- **C3 (Scientific Force Multiplication, proved in Section 3):** Under A5, AI has made beyond-record contributions in at least two independent scientific fields, documented a $> 2\times$ time speedup and a $> 100\times$ discovery-scale ratio, spanning four independent scientific disciplines.

- **C4 (Structural Unavoidability, proved in Section 4):** Under A3, A4, and A4b, AI is embedded in four critical societal systems with switching costs $S(d_X) \geq B_X$ (financially prohibitive); continued use only increases these costs; no viable non-AI substitute exists; AI's role is structurally unavoidable.

## Main Theorem

**Theorem 1** (AI Is the Future — Main Result)**.** *Under Assumptions A1, A2, A3, A4, A4b, and A5, Artificial Intelligence constitutes the defining technological paradigm of humanity's future development, as evidenced jointly by C1–C4:*

1. Technological Inevitability (C1): *AI capability growth is self-reinforcing, accelerating, and makes the pre-AI baseline economically dominated in every period.*

2. Economic Lock-In (C2): *At any integration depth $d \geq 1$, AI adoption is individually economically irreversible: no positive discount rate makes reversion rational.*

3. Scientific Force Multiplication (C3): *AI is a cross-domain scientific accelerant with demonstrated beyond-record contributions, quantified speedups, and global disciplinary reach.*

4. Structural Unavoidability (C4): *AI is already embedded in critical societal infrastructure at financially prohibitive switching-cost depth, with no viable substitute, making its continued role structurally unavoidable.*

*Together, C1–C4 establish a convergent, mutually reinforcing case: AI is not a transient technology but the defining paradigm of humanity's future.*

*Proof.* Each of C1, C2, C3, C4 is an independently proved conclusion:

- C1 is proved in Section 1 from A1 alone (Theorem 1 of Section 1).
- C2 is proved in Section 2 from A2 and A4 alone (Theorem 1 of Section 2).
- C3 is proved in Section 3 from A5 alone (Theorem 1 of Section 3).
- C4 is proved in Section 4 from A3, A4, and A4b (Theorem 1 of Section 4).

The dependency graph is acyclic: no conclusion is used to prove any other. All assumptions A1, A2, A3, A4, A4b, A5 are empirically documented and declared in the problem statement.

The four pillars are *convergent*: C1 shows AI's capability will keep improving (no technical plateau), C2 shows existing adopters cannot economically exit (lock-in), C3 shows AI accelerates the frontier of human knowledge (scientific compounding), and C4 shows AI is already embedded at the infrastructure level with no exit path. No single counter-argument can undercut all four simultaneously.

Therefore, under the stated assumptions, AI constitutes the defining technological paradigm of humanity's future development. □ □